

```

package uap.transport.app;

import uap.transport.*;
import java.util.Scanner;

public class TransportApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        TransportService service = new TransportService("UAP Transport");

        while (true) {
            System.out.println("\n\n===== UAP Transport Service =====");
            System.out.println("\n1. Add User");
            System.out.println("\n2. Increase Fee");
            System.out.println("\n3. View Cost");
            System.out.println("\n4. View Specific User");
            System.out.println("\n5. View All Users");
            System.out.println("\n6. Exit");

            int choice = sc.nextInt();

            switch (choice) {
                case 1:
                    System.out.println("1. Monthly 2. Per Trip");
                    int type = sc.nextInt();
                    sc.next();

                    System.out.print("Name: ");
                    String name = sc.next();
                    System.out.print("ID: ");
                    String id = sc.next();
                    System.out.print("Route: ");
                    String route = sc.next();

                    if (type == 1) {
                        System.out.print("Monthly Fee: ");
                        double fee = sc.nextDouble();
                        service.addUser(new MonthlyPassUser(name, id, route, fee));
                    } else {
                        System.out.print("Trip Rate: ");
                        double rate = sc.nextDouble();
                        System.out.print("Trip Count: ");
                        int count = sc.nextInt();
                        service.addUser(new PerTripUser(name, id, route, rate, count));
                    }
                    break;

                case 2:
                    System.out.print("Enter ID: ");
                    String id2 = sc.next();
                    System.out.print("Increase Amount: ");
                    double amt = sc.nextDouble();
                    service.increaseCost(id2, amt);
                    break;

                case 3:
                    System.out.print("Enter ID: ");
                    String id3 = sc.next();
                    System.out.println("Cost: " + service.getCost(id3));
                    break;

                case 4:
                    System.out.print("Enter ID: ");
                    String id4 = sc.next();
                    TransportUser u = service.findUser(id4);

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        if (u != null)
            System.out.println(u.toString(true));
        else
            System.out.println("User not found");
        break;

    case 5:
        for (TransportUser user : service.getAllUsers()) {
            System.out.println(user.toString(true));
        }
        break;

    case 6:
        System.exit(0);
    }
}
}
}
}

```

TransportService.java

```

package uap.transport;

import java.util.ArrayList;

public class TransportService{
    private String serviceName;
    private ArrayList<TransportUser> users;

    public TransportService(String serviceName){
        this.serviceName = serviceName;
        users = new ArrayList<>();
    }

    public void addUser(MonthlyPassUser user){
        users.add(user);
    }
    public void addUser(PerTripUser user) {
        users.add(user);
    }

    public TransportUser findUser(String id) {
        for (TransportUser u : users) {
            if (u.getId().equals(id)) {
                return u;
            }
        }
        return null;
    }

    public void increaseCost(String id, double amount) {
        TransportUser user = findUser(id);
        if (user != null) {
            user.increaseCost(amount);
        }
    }
    public double getCost(String id) {
        TransportUser user = findUser(id);
        if (user != null) {
            return user.getCost();
        }
        return 0;
    }
    public ArrayList<TransportUser> getUsers(boolean monthly) {
        ArrayList<TransportUser> result = new ArrayList<>();
    }
}

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        for (TransportUser u : users) {
            if (monthly && u instanceof MonthlyPassUser) {
                result.add(u);
            } else if (!monthly && u instanceof PerTripUser) {
                result.add(u);
            }
        }
        return result;
    }
    public ArrayList<TransportUser> getAllUsers() {
        return users;
    }

    public String getServiceName() {
        return serviceName;
    }
}

```

PerTripUser.java

```

package uap.transport;

public class PerTripUser extends TransportUser{
    private double tripRate;
    private int tripCountPerMonth;

    public PerTripUser(String name,String id,String route,double tripRate,int tripCountPerMonth){
        super(name,id,route);
        this.tripCountPerMonth = tripCountPerMonth;
        this.tripRate = tripRate;
    }

    public double getTripRate() {
        return tripRate;
    }

    public int getTripCountPerMonth() {
        return tripCountPerMonth;
    }

    public void setTripCountPerMonth(int tripCountPerMonth) {
        this.tripCountPerMonth = tripCountPerMonth;
    }

    public void setTripRate(double tripRate) {
        this.tripRate = tripRate;
    }
    @Override
    public void increaseCost(double amt){
        tripRate += amt;
    }
    @Override
    public double getCost(){
        return tripCountPerMonth*tripRate;
    }
    @Override
    public String toString(boolean details){
        if(!details){
            return super.toString();
        }else{
            return super.toString() + "; Rate: " + tripRate;
        }
    }
}

```

MonthlyPassUser.java

```
package uap.transport;

public class MonthlyPassUser extends TransportUser{
    private double monthlyFee;

    public MonthlyPassUser(String name,String id,String route,double monthlyFee){
        super(name,id,route);
        this.monthlyFee = monthlyFee;
    }

    @Override
    public void increaseCost(double amt){
        monthlyFee += amt;
    }
    @Override
    public double getCost(){
        return monthlyFee;
    }
    @Override
    public String toString(boolean details){
        if(!details){
            return super.toString();
        }else {
            return super.toString() + "; Fee: " + monthlyFee;
        }
    }
}
```

TransportUser.java

```
package uap.transport;

public abstract class TransportUser {
    private String name;
    private String id;
    private String route;

    public TransportUser(String name,String id,String route){
        this.name = name;
        this.id = id;
        this.route = route;
    }

    public String getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public String getRoute() {
        return route;
    }
    @Override
    public String toString(){
        return "Name: " + name + "; Id: " + id + "; Route: " + route;
    }
    public abstract double getCost();
    public abstract void increaseCost(double amt);
}
```

```
public abstract String toString(boolean details);  
}
```



```
Run TransportApp x  
"C:\Program Files\Java\jdk-25\bin\java.exe" "-javaagent:C:\Users\phara\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=58533" -Dfile.encoding=UTF-8 -Dse  
===== UAP Transport Service =====  
1. Add User  
2. Increase Fee  
3. View Cost  
4. View Specific User  
5. View All Users  
6. Exit
```